MARCO ANGELO P. QUANICO

AI Agent Developer | Agentic AI, RAG, Local LLMs & Computer Vision

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PROFESSIONAL SUMMARY

AI/ML and software engineer with approximately 1.5 years of professional engineering experience across healthcare IT systems and AI-native business systems, backed by an MS in Computer Science (in progress) and published computer vision research. Hands-on with Python, Laravel, PostgreSQL/pgvector, RAG, multi-agent orchestration, tool/function calling, prompt compression, local LLM deployment (Ollama), and YOLO-based vision pipelines. Experienced owning AI workflows from schema design and knowledge-base integration through background orchestration, token-cost optimization, and production support, including legal-services and hospital environments.

TECHNICAL SKILLS

Agentic AI	Agentic Orchestration, Multi-Agent Workflows, Tool / Function Calling, Prompt Engineering & Compression, Context Window Optimization, n8n Automation
LLM / RAG	RAG, Vector Databases, pgvector, LlamaIndex, Ollama (Local LLM Deployment), Llama 3.1, Qwen2.5-Coder / Qwen2.5-VL, CodeGemma, GPT-OSS, GPU Inference (RTX 3060)
Computer Vision	YOLOv8, YOLOv11 (Detection, Instance Segmentation), VLM-based OCR, Re-Identification Tracking, CVAT, Jupyter Notebooks
Engineering	Python, PHP (Laravel, Livewire, Filament), JavaScript / Node.js, React / TypeScript, PostgreSQL, MSSQL, Redis, Docker, REST APIs, Git, Kotlin, Java, Rust

PROFESSIONAL EXPERIENCE

AI-Native Systems Engineer / Developer

Rosian Liaison Services | Mar 2026 – Present | Bacolod City, Philippines

- Architected AI-native workflows that orchestrate 5 specialized agent skills with knowledge-base integration (RAG), optimizing context usage and minimizing API round-trips per task.
- Accelerated task execution by 70% and expanded inquiry-processing throughput through background AI orchestration, reducing manual workload on repetitive daily operations by 80%.
- Cut operational expenses by 33%, lowering department overhead from ₱15,000 to ₱10,000 per month by absorbing the throughput of a full-time staff role without headcount replacement.
- Tracked and optimized token economics, averaging 89K tokens per run on gpt-6-luna ($0.20/MTok), or about $0.0178 (₱1.11) per completed task.
- Designed a scalable PostgreSQL schema for vector embeddings, knowledge bases, and multi-agent interaction logs.
- Built solutions for a liaison business supporting attorneys, legal professionals, and real estate stakeholders using PHP/Laravel (Livewire, Filament, Alpine), React/TypeScript, Tailwind, Redis, and PostgreSQL; optimized legal transaction workflows through process automation, improving task completion efficiency by 10%.
- Conducted usability observation sessions in production use to identify workflow bottlenecks and friction points. Recognized as Best Staff of Q2 and Q3 2026 for operational savings and AI workflow integration.

AI Systems Developer (Project-Based Contract)

Rosian Liaison Services | Jan 2026 – Mar 2026 | Bacolod City, Philippines

- Built an offline AI-assisted legal document generation system on local LLMs (Llama 3.1:8B, CodeGemma, Qwen2.5-Coder) running on an RTX 3060, eliminating cloud reliance and typist errors.
- Implemented RAG with a vector database for contextual retrieval and prompt compression, and built a Qwen2.5-VL pipeline that extracts structured text from hardcopy records into retrieval workflows.
- Engineered HTML template parsing that preserves text styling, achieving 98% accuracy in variable extraction and 75% precision in complex styling conversion.

System Developer (Probationary)

Adventist Medical Center Bacolod | Jun 2025 – Jan 2026 | Bacolod City, Philippines

- Led maintenance and continuous improvement of a custom HR information system (C#/ASP.NET WinForms, MSSQL), resolving production issues and clearing 60% of the outstanding fix backlog.
- Architected and implemented a shift scheduling module with shift-swap and approval workflows and the supporting MSSQL schema, reducing manual scheduling overhead and improving internal coordination.
- Consulted HR stakeholders to gather requirements and translate operational needs into system enhancements; provided end-to-end IT support, including ticket resolution and network and hardware troubleshooting for hospital staff.
- Contributed to hospital-wide research initiatives and was recognized as an ICare Champion for Q3 2025 for service excellence.

Software Developer (Practicum / Intensive Training)
Riverside Medical Center Inc. | Mar 2025 – Jun 2025 | Bacolod City, Philippines

- Designed and developed the institution's doctor directory web application (PHP/Laravel, Livewire, PostgreSQL), including the schema for search, filtering, and record management.
- Collaborated with the institution's engineers through iterative feedback cycles, achieving 85% UX satisfaction while refining the database schema from technical review, and practiced version control and code review in a live institutional environment.

SELECTED PROJECTS

- Agentic AI (Cokeeter): Built an agentic system of specialized AI agents for database administration, security, secretarial tasks, and statistical analysis, orchestrated to handle work comparable to a small business workforce, using Laravel 12, Redis 7.2, PostgreSQL 17, and Ollama (local/cloud).
- AI-Native Medical Doctors Platform (MABDOC): Lead developer of a full-stack clinic platform (React, Laravel, REST API, PostgreSQL) with doctor scheduling, directory, real-time notifications, and role-based access control; added queue and scheduled jobs, email notifications, and Cloudflare R2 storage, deployed on Ubuntu with CI/CD and Grafana monitoring.
- n8n Workflow Automations: Built a multi-label Gmail-to-Sheets & Drive automation and a Telegram receipt-photo workflow that extracts and logs receipt data automatically.

AI RESEARCH

- AI-Driven YOLOv11 Instance Segmentation Model for Automated Seashell Detection, Classification and Harvest Assessment in Coastal Communities of Negros Occidental. Co-author; published Sept 2026 (ISTES).
- PPE and Hand-Hygiene Monitoring Using Multi-Location CCTV and YOLO-Based Re-Identification Tracking. Author; MSCS thesis concept preparation (on-going).
- IoT-Based Red Tide Extraction, Assessment, and Detection Classification Using AI-Driven YOLOv11. Co-author; on-going.
- Determining Seashell Diversity Through Object Recognition Using the YOLOv8 Model. Author; BSCS thesis (Dec 2024).

EDUCATION

Master of Science in Computer Science | Recoletos de Bacolod Graduate School | 2025 – Present
9 units completed; on-going

Bachelor of Science in Computer Science | STI West Negros University | 2021 – 2025
Best in Thesis A.Y. 2025; Champion, Best Thesis, 19th ICT EXPO 2025; Champion, ICT Olympics 2024 (Web Design and Development); Top 2 Best CS Project Manager, 19th ICT EXPO 2025; 1st Runner-Up, 2025 Codefest